EchoTrack: Product Requirements Document

Version 1.0 · November 4, 2026 · Status: v1 shipped as a web app, v2 proposed

1. Summary

EchoTrack is a lightweight daily sustainability logger. In under a minute, a person records how they travelled, what they ate, how much energy they used at home, and which small eco-habits they kept. The app turns those choices into an estimated daily carbon footprint (kg CO_2e), compares it with a personal target, and shows a 7-day trend and logging streak.

2. Problem and opportunity

Most people who want to live more sustainably have no feedback loop. Carbon calculators are built for one-off annual estimates, require long forms, and don't reward daily consistency. Habit trackers build consistency but say nothing about impact. EchoTrack sits between the two: fast daily entry, immediate impact feedback, and gentle streak-based motivation.

3. Goals and non-goals

Goals

- Make a complete daily log possible in under 60 seconds.
- Give instant, understandable feedback on the day's footprint versus a target.
- Encourage consistency through streaks and a weekly view.
- Work offline-friendly, with no account and no data leaving the device.

Non-goals for v1

- Lifecycle-accurate or audited carbon accounting.
- Social features, leaderboards or carbon offsetting.
- Automatic data capture (GPS, smart meters, bank data).
- Multi-user or household profiles.

4. Target users

- **Curious starters:** want to understand which daily choices matter most, with no jargon.
- **Habit builders:** already trying to cut their footprint and want a streak and trend to keep going.
- **Students and teams running challenges:** want a simple shared language for comparing days, once sharing exists (v2).

5. User stories

1. As a user, I log today's travel, food and home energy by tapping one option per category, so that logging stays quick.
2. As a user, I tick small wins such as a reusable bag or sorted waste, so that good habits visibly reduce my total.
3. As a user, I see my total against a daily target, so that I know whether the day went well.
4. As a user, I review the last 7 days, my average and my streak, so that I can see progress.
5. As a user, I can go back to a previous day to log or correct it, so that a missed day doesn't break my record.
6. As a user, I can add a short note, so that I remember what I want to change tomorrow.
7. As a user, I can clear a day, so that mistakes are easy to undo.

6. Functional requirements (v1, shipped)

ID	Requirement	Priority
F1	Three single-choice categories: getting around, food, energy at home. Tapping a selected option deselects it.	Must
F2	Six toggle habits, each subtracting a fixed amount from the total.	Must
F3	Live daily total in kg CO_2e, never below zero.	Must
F4	Meter and plain-language verdict against a 6 kg/day target.	Must

ID	Requirement	Priority
F5	Date picker with previous and next buttons. Future dates are blocked.	Must
F6	7-day bar chart ending on the selected date, with the target line.	Must
F7	Weekly average, consecutive-day logging streak, and days under target.	Should
F8	Free-text daily note.	Should
F9	Clear-day action.	Should
F10	Local persistence in the browser, wrapped in error handling so the app works if storage is unavailable.	Must
F11	Light and dark themes following the system setting.	Should

7. Footprint model

The model uses fixed, rounded estimates per choice. They are intentionally simple and are shown to users as rough figures.

Category	Options and estimated kg CO_2e
Getting around	Walk or cycle 0 · Worked from home 0 · Public transit 0.9 · Two-wheeler 1.1 · Carpool 1.6 · Drove alone 3.2
Food	Plant-based 1.5 · Vegetarian 2.5 · Some meat or fish 4.5 · Beef or lamb 7.0
Energy at home	Low use 1.5 · Typical 3.0 · AC for hours 5.5
Habits (subtract)	Reusable bag or bottle 0.2 · Sorted waste 0.3 · Composted scraps 0.3 · No single-use plastic 0.2 · Line-dried clothes 0.5 · Short shower 0.2

Daily total = sum of selected category values minus selected habit credits, floored at zero. A day counts as logged once any option is chosen. The default target is 6 kg/day.

8. UX principles

- One tap per decision. No forms, no typing required.
- Plain language over carbon jargon.
- Feedback is encouraging, never shaming: an over-target day suggests a next step.
- Mobile first, with visible keyboard focus and readable contrast in both themes.

9. Success metrics

Metric	Target
Median time to complete a daily log	Under 60 seconds
Share of active users who log 4 or more days in a week	40%
Seven-day return rate	35%
Average streak length among returning users	5 days or more
Week-over-week reduction in average daily footprint for users active 4 weeks	10%

v1 stores data only on the device, so these metrics need opt-in analytics or a hosted backend (see v2).

10. Technical notes

- Single self-contained HTML page: inline CSS and JavaScript, one web font with a system fallback, inline SVG chart.
- Data stored in browser localStorage under one key, keyed by date (YYYY-MM-DD). Storage is per device and per browser.
- No network requests beyond the font stylesheet. No accounts, no tracking.

11. Risks and open questions

- **Accuracy:** fixed estimates ignore region, grid mix, distance and portion size. Mitigation: label figures as estimates and add region presets in v2.

- **Fixed target:** 6 kg/day may be unrealistic for some users. Open question: should the target be editable in v1.1?
- **Data loss:** clearing browser data erases history. Mitigation: export and import in v1.1.
- **Gaming and guilt:** a streak can discourage users after a missed day. Consider a “rest day” allowance.

12. Roadmap

v1.1

- Editable daily target.
- Export and import of data as JSON or CSV.
- Rest-day allowance that preserves streaks.

v2

- Region-aware factors, for example Indian grid electricity and local transit modes.
- Distance and portion inputs for finer estimates.
- Weekly and monthly reports with the biggest-impact category highlighted.
- Optional sync across devices and opt-in analytics.
- Shared challenges for classes, teams and families.

Later

- Reminders and notifications.
- Integrations with calendars and transit apps for suggested entries.

13. Out of scope

Carbon offsets, marketplace features, financial incentives, and any claim of certified emissions reporting.